ALEXANDER PANCHENKO

UNREAL ENGINE C++ GAMEPLAY PROGRAMMER

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LinkedIn: https://www.linkedin.com/in/alexander-panchenko-0b4b47414
GitHub: Tactical Turn-Based Combat | Targeted Limb Combat
Project YouTube demos: Tactical Turn-Based Combat | Targeted Limb Combat

PROFILE

Unreal Engine C++ gameplay programmer with professional UE5 experience and a BSc in Computer Science. Experienced in gameplay systems, GAS, AI, replication, and custom pathfinding algorithms, with a focus on C++ architecture and gameplay logic.

TECHNICAL SKILLS

Core: C++ | Unreal Engine 5 | GAS
Features: Gameplay Tags | FastArraySerializer | Pathfinding | Inventory systems | Behavior Trees | Enhanced Input | UI
Tools & Practices: Git | Perforce | JetBrains Rider | Visual Studio | Component-based design | Data-driven systems | Debugging | Profiling

PROFESSIONAL EXPERIENCE

Unreal Engine Game Programmer | Delta Design - Unannounced Project | Nov 2024 - Jun 2026
- Designed, implemented, and scaled modular GAS-based gameplay abilities, establishing robust state transitions, predictive activation flows, and Gameplay Tag-driven rulesets to support complex combat mechanics
- Architected and developed high-performance C++ HUD components tightly coupled with gameplay events, minimizing Blueprint overhead and ensuring low lag UI updates during state changes .
- Engineered a robust player movement and Enhanced Input pipeline, decoupling control inputs from core character logic and ensuring seamless synchronization with existing systems .
- Investigated, profiled, and resolved critical gameplay bugs and race conditions involving ability execution and multi-system cross-integration .
- Collaborated with the team to integrate features into the existing codebase, enforcing strict C++ architectural patterns, clean component-based design, and long-term code maintainability.

SELECTED PROJECTS

TACTICAL TURN-BASED COMBAT SYSTEM | Unreal Engine 5, C++
- Implemented tactical grid movement with movement-point budgeting and collision-aware reachability; integrated custom JPS pathfinding with runtime updates for dynamic obstacles and generated reachable-cell visuals.
- Implemented initiative-based turn scheduling with participant order, team sequencing, and action-point replenishment.
- Built data-driven GAS abilities for melee, self-target, and projectile actions; configured targeting, range, area shape, Gameplay Tags, and action-point costs, and added XP-based leveling with Tag-gated skill-tree requirements.
- Built a replicated inventory using FFastArraySerializer and server RPCs for item add, drop, consume, and equip operations.
- Implemented spatial-grid inventory with stack handling and splitting, category filters, and character-specific equipment grids; fragment-based item manifests, attached equipped actors to character meshes, and connected pickup and weapon-stat flows.
- Implemented Behavior Tree target search and melee execution, filtering by team, range, and action points before moving into attack position.
- Developed combat HUD widgets for turn order, action points, movement range, and character attributes.

TARGETED LIMB COMBAT SYSTEM | Unreal Engine 5, C++
- Implemented tag-driven melee hit detection using swept sphere traces between weapon sockets; mapped hit bones to limb Gameplay Tags and calculated damage with limb-specific armor and multipliers via custom Gameplay Effect Execution Calculation.
- Connected combo attacks and dodge abilities to animation montages and gameplay events.
- Projected limb positions into screen-space target widgets and broadcast limb-health changes to UI listeners.
- Added limb-destruction feedback by hiding damaged bones, attaching wound meshes, and spawning physics-driven detached limb actors.

BLOCKCHAIN LAND REGISTRY | Bachelor thesis, C++
- Built a local-network peer-to-peer prototype for a cadastral registry, implementing node communication, transaction validation, and block synchronization.

EDUCATION

Bachelor of Computer Science | Taras Shevchenko National University of Kyiv
Specialization: Mathematical Informatics | Thesis: Application of Blockchain Technologies in Land Cadastre Registration